Solmaz Haddady, MSc

AI Researcher & Machine Learning Engineer

Computer Vision · Deep Learning · Multimodal AI · Medical Data Engineering

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PROFESSIONAL PROFILE

AI Researcher and Machine Learning Engineer with an MSc in Artificial Intelligence and a strong quantitative background in bioinformatics and physics. Hands-on experience developing end-to-end AI pipelines, deep-learning models, and multimodal vision–language systems, from data preprocessing and quality control to model evaluation. Developed CNN–Transformer models for medical image classification and a vision–language pipeline for automated histopathology report generation, working with 3,000 whole-slide images. Experienced in Python-based AI development, scientific data analysis, and interdisciplinary collaboration, with transferable skills for applied AI and data-driven engineering across research and industry.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer & Medical AI Researcher

Department of Dermatology and Venereology, JKU Linz & KUK (Kepler University Hospital) | 2023 – 2025

Data Engineering & Dataset Development

- Built a curated dataset of 3,000 whole-slide images (WSI): retrieved slides from the pathology archive, cleaned, digitized and quality-checked them.
- Labelled slides with pathologists and cross-validated every label against the original patient reports to ensure label reliability.
- Anonymised images and records to protect patient privacy, designed structured metadata (pseudonymized patient ID, main diagnosis, sub-diagnosis).
- Converted proprietary WSI formats (SVS, MRXS) to DICOM, enabling standardized, interoperable and resource-efficient data handling for ML.

Research Contributions

- Prepared the dataset for public release on The Cancer Imaging Archive (TCIA).

AI Engineering & ML Development

- Designed CNN–Transformer models for WSI classification and rare-pattern recognition (96.65% validation accuracy).
- Developed a multimodal vision–language pipeline (LLaMA, LoRA) linking WSI representations with pathology report text for automated report generation.
- Evaluated models with statistical validation and benchmarking, focusing on robustness and reproducibility.
- Collaborated daily with dermatologists, pathologists and clinical researchers.

AI Research Assistant – Digital Pathology

Department of Pathology & Molecular Pathology, JKU Linz | 03/2023 –10/2023

- Contributed to developing AI models for **computer-assisted diagnosis and tumor regression grading** in digital pathology.
- Prepared, processed, and validated clinical imaging datasets for machine-learning workflows in collaboration with AI and medical researchers.

Seismic Data Analysis & Data Quality

OEOC Co., Tehran, Iran | 2014 – 2017

- Processed and analysed seismic time-series data using signal processing and time–frequency analysis for geological interpretation and gas-detection research.
- Applied numerical modelling, simulation, preprocessing, and noise-reduction techniques to investigate signal patterns.
- Performed data quality checks, consistency validation, and metadata standardization to improve the reliability and usability of scientific datasets.

Tutor for Physics & Mathematics

Self-Employed, Tehran, Iran | 2014 – 2017

- Instructed university and high school students in mathematics and physics.

SELECTED AI PROJECTS

DermPath-LLaMA — Multimodal Histopathology Report Generation

Python | PyTorch | Transformers | LoRA | LLaMA

- Developed a multimodal vision–language pipeline integrating whole-slide image representations with pathology report text for automated report generation.
- **Repository:** https://github.com/solmazhaddady/DermPath_LLaMA_VLM

NMSC Whole-Slide Imaging Dataset

Python | WSI | DICOM | OpenSlide | Pandas

- Curated and standardized approximately 3,000 whole-slide images and associated metadata, including preprocessing, quality control, and anonymisation workflows for AI research and dataset preparation.
- **Repository:** https://github.com/solmazhaddady/NMSC-TCIA-Dataset

WSI-to-DICOM Converter — Medical Imaging Interoperability

Python | DICOM | OpenSlide | wsidicomizer

- Developed an automated pipeline to convert proprietary whole-slide image formats (SVS, MRXS) into DICOM for standardised medical-image handling.
- **Repository:** github.com/solmazhaddady/WSI-to-DICOM-Converter

Whole-Slide Image Quality Control Pipeline

Python | Computer Vision | Image Processing

- Developed image-validation and quality-control workflows to identify data-quality issues and prepare reliable datasets for machine-learning pipelines.

EDUCATION

Master of Science in Artificial Intelligence

Johannes Kepler University (JKU), Linz, Austria | 2022 – 2025

- Specialization: Machine Learning, Deep Learning, Computer Vision, Medical Imaging
- Thesis: Classification and Vision–Language Modelling for Skin Cancer Histopathology (Grade: excellent)

- Presented research at the **9th & 10th ÖGDV Science Days (2024 & 2025), Austria**

Bachelor of Science in Bioinformatics
Johannes Kepler University (JKU), Linz, Austria | 2017 – 2022

- Interdisciplinary studies combining Computer Science and Molecular Biology
- Erasmus+ Exchange: University of South Bohemia, Czech Republic.

Master of Science in Geophysics
Tehran University, Iran | 2015 – 2017

- Focus on numerical modelling, wave physics, and earth sciences.
- Thesis: Gas and low-frequency shadows detection with the S-transform time-frequency method (Grade: excellent)

Bachelor of Science in Physics
Tehran University, Iran | 2012 – 2015

- Foundational studies in theoretical physics and mathematics.

RESEARCH EXPERIENCE

- **International Summer School on Seismology, Shiraz University, Iran | 2016**
- **Poster Presentation – 9th ÖGDV Science Days, Bad Gastein, Austria (2024)**
- **Poster Presentation – 10th ÖGDV Science Days, Werfenweng, Austria (2025)**

TECHNICAL SKILLS

- Programming & Data Processing: Python, R, Pandas, NumPy, scikit-learn
- Machine Learning & Deep Learning: PyTorch, TensorFlow, CNNs, Transformers, Vision Transformers, image classification, feature extraction, model evaluation, benchmarking
- Computer Vision & Medical AI: Computer vision, image processing, whole-slide imaging (WSI), digital pathology, medical image analysis, DICOM, OpenSlide, wsidicomizer
- Multimodal & Generative AI: Large Language Models (LLMs), Vision–Language Models (VLMs), multimodal learning, LoRA, automated report generation
- Data Engineering & Computing: Data preprocessing, dataset curation, metadata validation, data standardisation, quality control, Git, Linux, Docker, CUDA/GPU computing, Jupyter, PyCharm, VS Code

Engineering Principles

Data privacy by design · label quality through expert review · standardised formats (DICOM) · reproducible, resource-efficient pipelines

SELECTED PUBLICATION & MANUSCRIPTS

Haddady, S., *et al.* (Manuscript in preparation).
"A High-Resolution Dataset of Non-Melanoma Skin Cancer Whole-Slide Images for Machine Learning Research" — planned submission to *Scientific Data* (Nature Portfolio).

LANGUAGES

- English: Professional Working Proficiency
- German: B2
- Persian: Native